

DATA PREPARATION I

SESSION 4
COMP6140001 – DATA MINING

SUBJECT MATTER EXPERT
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LEARNING OBJECTIVE

L01 : Explain concept of data, data warehouse, data preprocessing, and data mining



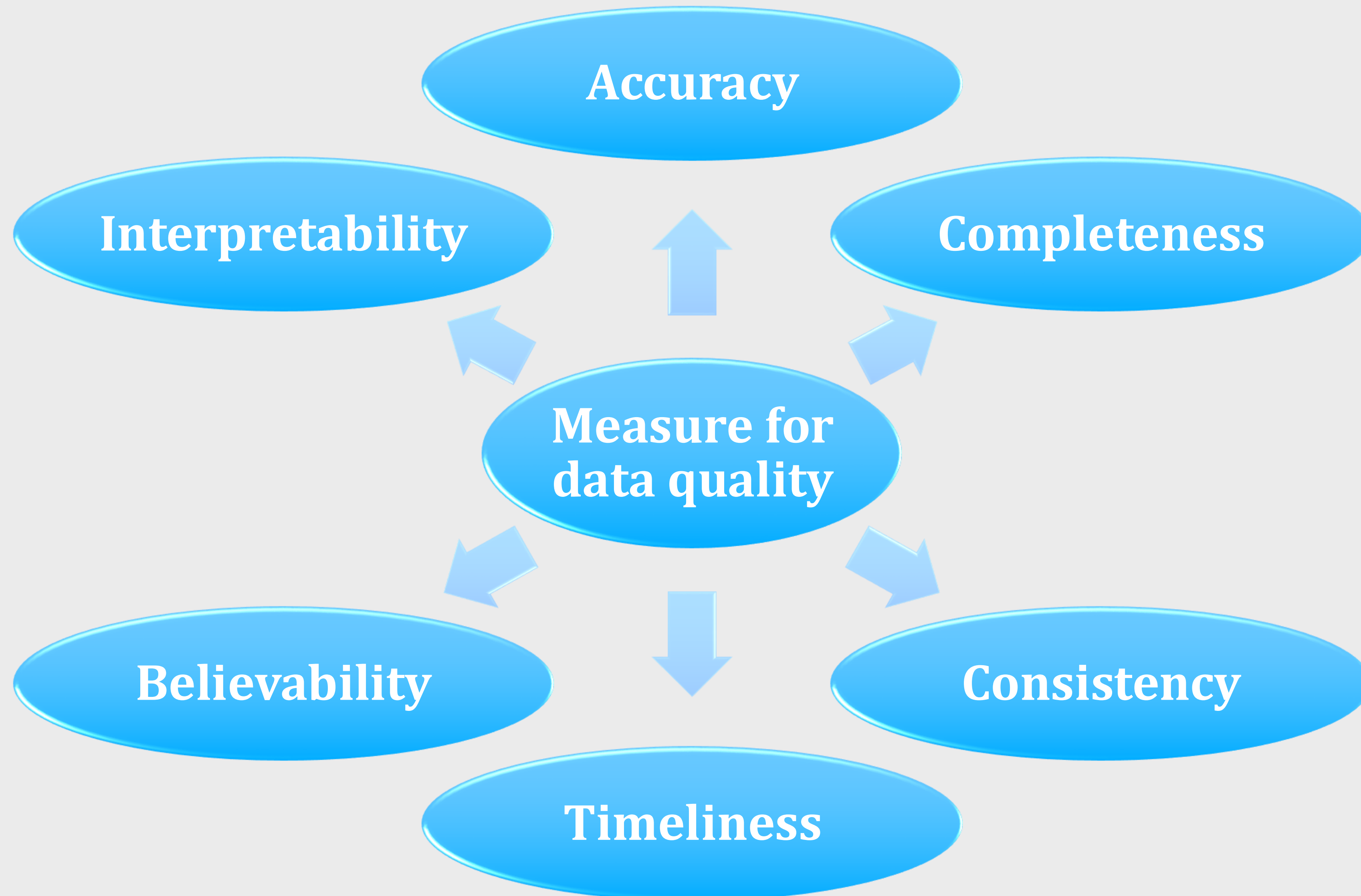
OUTLINE

- Data Quality Measure
- Data Cleaning
- Data Integration

DATA QUALITY MEASURE

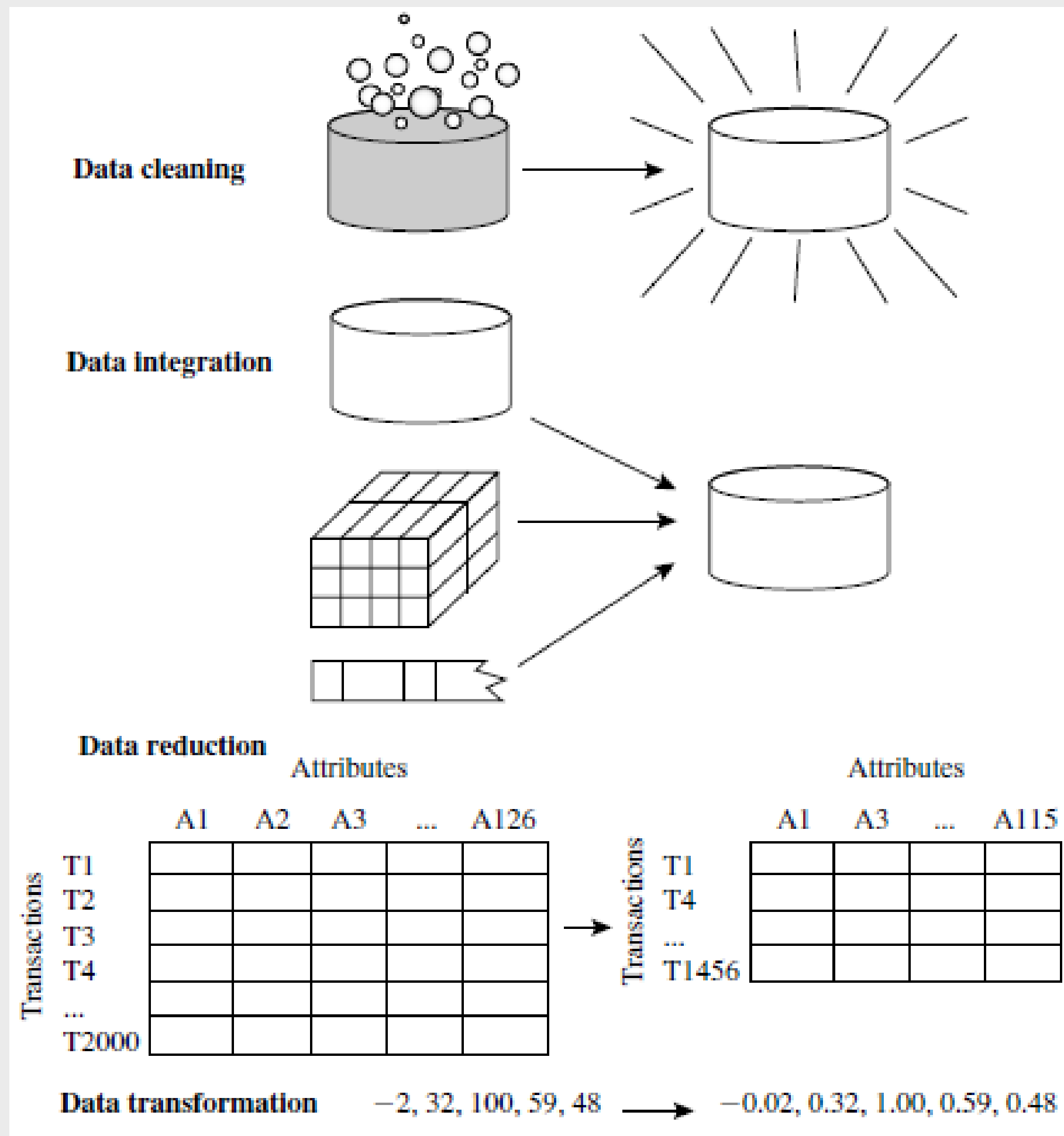


DATA QUALITY ISSUES





MAJOR TASKS IN DATA PREPROCESSING



- **Data cleaning:** Handle missing data, smooth noisy data, identify or remove outliers, and resolve inconsistencies
- **Data integration:** Integration of multiple databases, data cubes, or files
- **Data reduction:** Dimensionality reduction, Numerosity reduction, Data compression
- **Data transformation:** Normalization, Concept hierarchy generation

DATA CLEANING



DATA CLEANING

- Real world data is dirty: lots of potentially incorrect data, e.g., instrument faulty, human or computer error, transmission error
 - Incomplete (missing values): lacking attribute values, lacking certain attributes of interest, or containing only aggregate data, e.g., Occupation=" "
 - Noisy: containing noise, errors, or outliers, e.g., Salary="-10" (an error)
 - Inconsistent: containing discrepancies in codes or names, e.g.,
 - Age="42", Birthday="03/07/2010"
 - Was rating "1, 2, 3", now rating "A, B, C"
 - discrepancy between duplicate records
 - Intentional, e.g., disguised missing data



DATA CLEANING : MISSING VALUES

Incomplete (Missing) Values

- Data is not always available, e.g.: many tuples have no recorded value for several attributes, such as customer income in sales data
- Missing data may be due to
 - Equipment malfunction
 - Inconsistent with other recorded data and thus deleted
 - Data not entered due to misunderstanding
 - Certain data may not be considered important at the time of entry
 - Not register history or changes of the data
- Missing data may need to be inferred



DATA CLEANING : HANDLE MISSING VALUES

- Handle the missing data

Ignore the tuple

- usually done when class label is missing (when doing classification)—not effective when the % of missing values per attribute varies considerably

Fill in the missing value

- manually
- automatically with a global constant: e.g., “unknown” or $-\infty$
- automatically with a measure of central tendency, e.g., mean or median
- automatically with the attribute mean or median for all samples belonging to the same class as the given tuple
- automatically with **the most probable value**: inference-based such as Bayesian formula or decision tree



DATA CLEANING : NOISY DATA

- Noisy Data: A random error or variance in a measured variable
- Incorrect attribute values may be due to
 - Faulty data collection instruments
 - Data entry problems
 - Data transmission problems
 - Technology limitation
 - Inconsistency in naming convention
- Other data problems: duplicate records, incomplete, and inconsistent data



DATA CLEANING : HANDLE NOISY DATA

- Handle the noisy data

Binning

- First, sort the data and distribute into (equal-frequency) bins
- Then, one can **smooth by bin means, median, boundaries**, etc.

Regression

- Smooth by fitting the data into regression functions

Clustering

- Detect and remove outliers

Combined computer and human inspection

- Detect suspicious values and check by human (e.g., deal with possible outliers)



DATA CLEANING : DISCRETIZATION BY BINNING

- Top-down split, unsupervised method
- **Equal-width** (distance) partitioning
 - Divides the range into N intervals of equal size: uniform grid
 - If A and B are the lowest and highest values of the attribute, the width of intervals will be: $W = (B - A) / N$
 - The most straightforward, but outliers may dominate presentation
 - Skewed data is not handled well
- **Equal-depth** (frequency) partitioning
 - Divides the range into N intervals, each containing approximately same number of samples
 - Good data scaling
 - Managing categorical attributes can be tricky



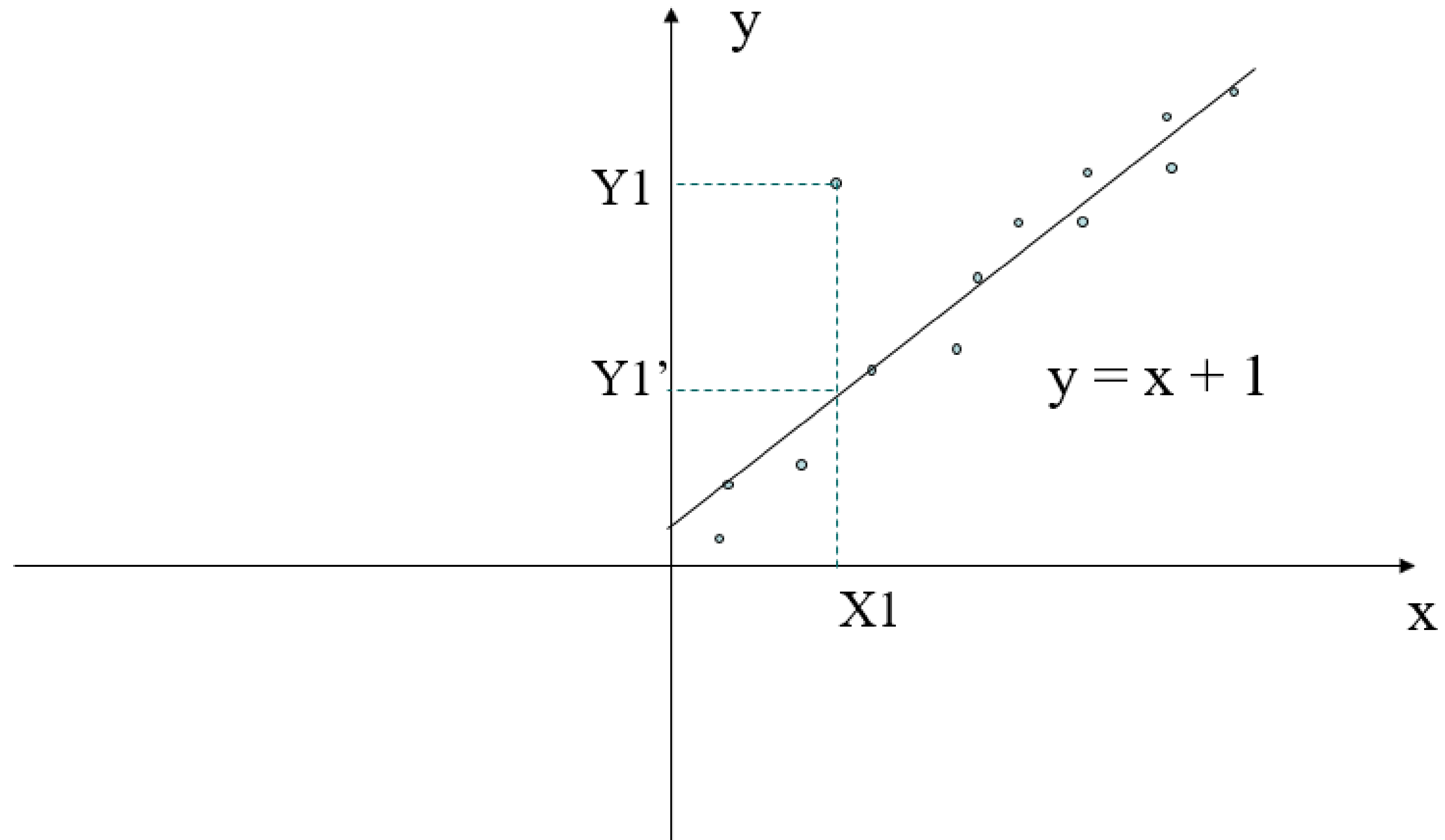
DATA CLEANING : EXAMPLE BINNING

- Example for data smoothing using binning method:
- Sorted data for price (in dollars): 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

Method	Partition into Equal-depth bins	Smoothing by bin means	Smoothing by bin boundaries
Bin			
1	4, 8, 9, 15	9, 9, 9, 9	4, 4, 4, 15
2	21, 21, 24, 25	23, 23, 23, 23	21, 21, 25, 25
3	26, 28, 29, 34	29, 29, 29, 29	26, 26, 26, 34



DATA CLEANING : REGRESSION





DATA CLEANING : AS A PROCESS

- Data Cleaning as a Process

Data discrepancy detection

- Use metadata (e.g., domain, range, dependency, distribution)
- Check field overloading
- Check uniqueness, consecutive, and null rule
- Use commercial tools:
 - Data scrubbing: use simple domain knowledge (e.g., postal code, spell-check) to detect errors and make corrections
 - Data auditing: by analyzing data to discover rules and relationship to detect violators (e.g., correlation and clustering to find outliers)

Data migration and integration

- Data migration tools: allow transformations to be specified
- ETL (Extraction/Transformation/Loading) tools: allow users to specify transformations through a graphical user interface

- Integration of the two processes:
- Iterative and interactive (e.g., Potter's Wheels)

DATA INTEGRATION



DATA INTEGRATION

- **Data** integration
 - Combines data from multiple sources into a coherent store
- **Schema** integration
 - Integrate metadata from different sources, e.g., A.cust-id $\square$ B.cust-#
- Entity identification problem:
 - Identify real world entities from multiple data sources, e.g., Bill Clinton = William Clinton
- Detecting and resolving data value conflicts
 - For the same real world entity, attribute values from different sources are different
 - Possible reasons: different representations, different scales, e.g., metric vs. British units



DATA INTEGRATION : REDUNDANCY

Data Redundancy

- Often occurs when integration of multiple databases
 - Object identification: the same attribute or object may have different names in different databases
 - Derivable data: one attribute may be a “derived” attribute in another table, e.g., annual revenue
- May be able to be detected by **correlation analysis** and **covariance analysis**
- Careful integration of the data from multiple sources may help reduce/avoid redundancies and inconsistencies and improve mining speed and quality

SUMMARY

- Measures for data quality: Accuracy, Completeness, Consistency
Timeliness, Believability, Interpretability
- Task in Data Preprocessing: Data cleaning, Data integration, Data
reduction, Data transformation

EXERCISE

1. Which one of the following options that is **NOT** issues in data quality?
 - a. Accuracy
 - b. Completeness
 - c. Efficiency
 - d. Interpretability
2. In a customer dataset, some records contain extreme values in the 'Age' column, such as 150 or -5. What technique would you recommend to handle these outliers, and why?
3. How to handle the missing values while data cleaning?

REFERENCES

Han, J., Kamber, M., & Pei, J. (2023). Data Mining: Concepts and Techniques (4rd ed.). San Francisco, CA, USA: Morgan Kaufmann Publishers Inc.